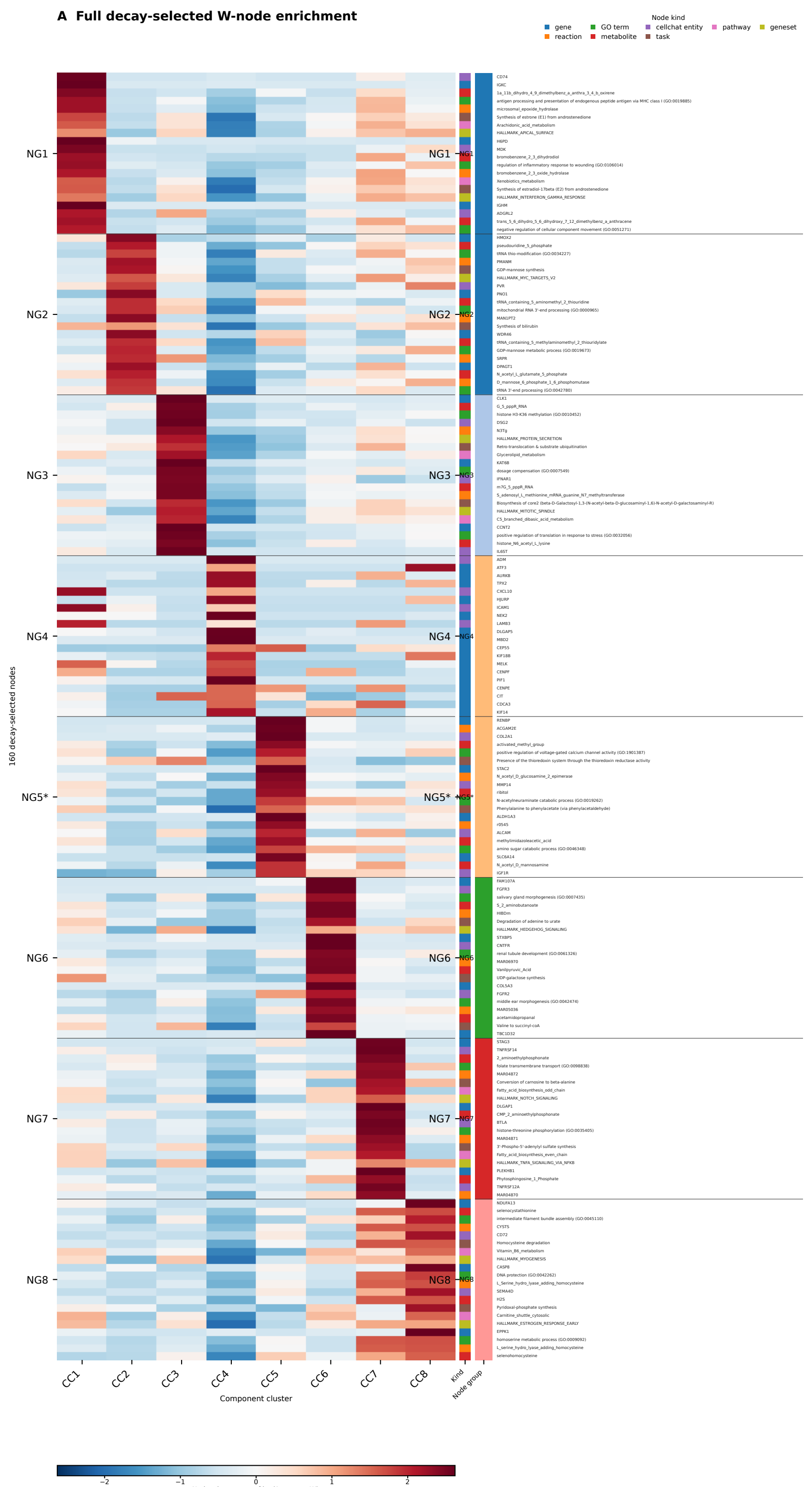A Full decay-selected W-node enrichment